

# ARM2SID Setup for MB6582

## Background

With the availability of genuine MOS SID Chips becoming scarcer and scarcer, it makes sense to look for modern and readily available alternatives for most (maybe even all) of the use cases where SID chips have traditionally been used, and MIDIBox synths are no exception.

We all *love* the fact that the MIDIBox SammichSID and MB6582 Synths provide support for multiple sound chips, which gives them their rich & multi-layered soundscapes, but finding and sourcing those 40yr antiques is risky, because there are lots of fakes and duds out there, as well as expensive (e.g., rates between \$50-\$100 USD per chip).

Fortunately there are plenty of SID Clone options out there, ranging from the SwinSID and SwinSID Ultimate, to the FPGA SID, BackSID, SIDKick, X-SID, and many others, but my personal favorite is the Nobomi ARMSID, available from <https://Retrocomp.cz>

For my taste, it provides very accurate sound reproduction, it has a clean design aesthetic, which looks great, as well as a form factor that's really close to the original MOS SID, which helps it fits everywhere a SID would. It's also reasonably priced at around ~\$30 USD, and it has a well maintained firmware codebase, which is regularly updated.

To that last point regarding firmware maintenance, some music synth owners were complaining about having to keep a working Commodore 64 around in order to run a program to update their ARMSID firmware. So, the Retrocomp team responded by creating the ARMSID Shield, which makes it super easy to update firmware, manage settings and configurations and even test sound reproduction, using real SID music.

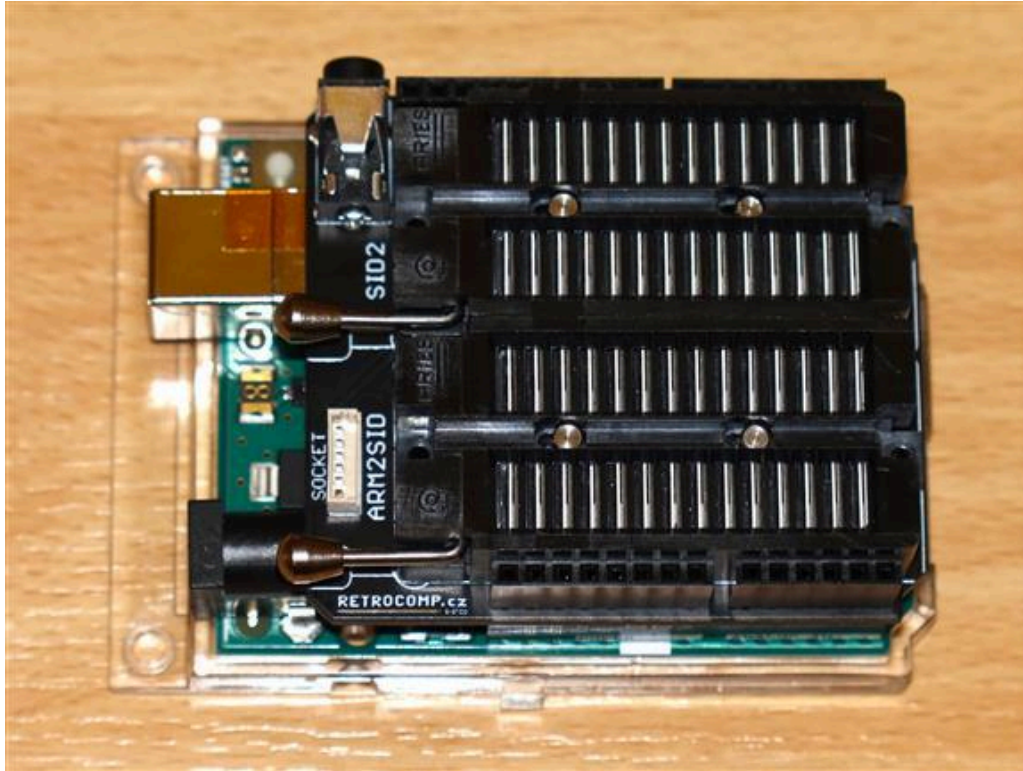
You can find plenty of info about the ARMSID and its Arduino shield on the Retrocomp.cz website.

The purpose of this guide is to help you get up and running quickly with your Arduino ARMSID Shield, so you can properly prepare your ARMSIDs and ARM2SIDs before placing them in your MIDIBox Synth

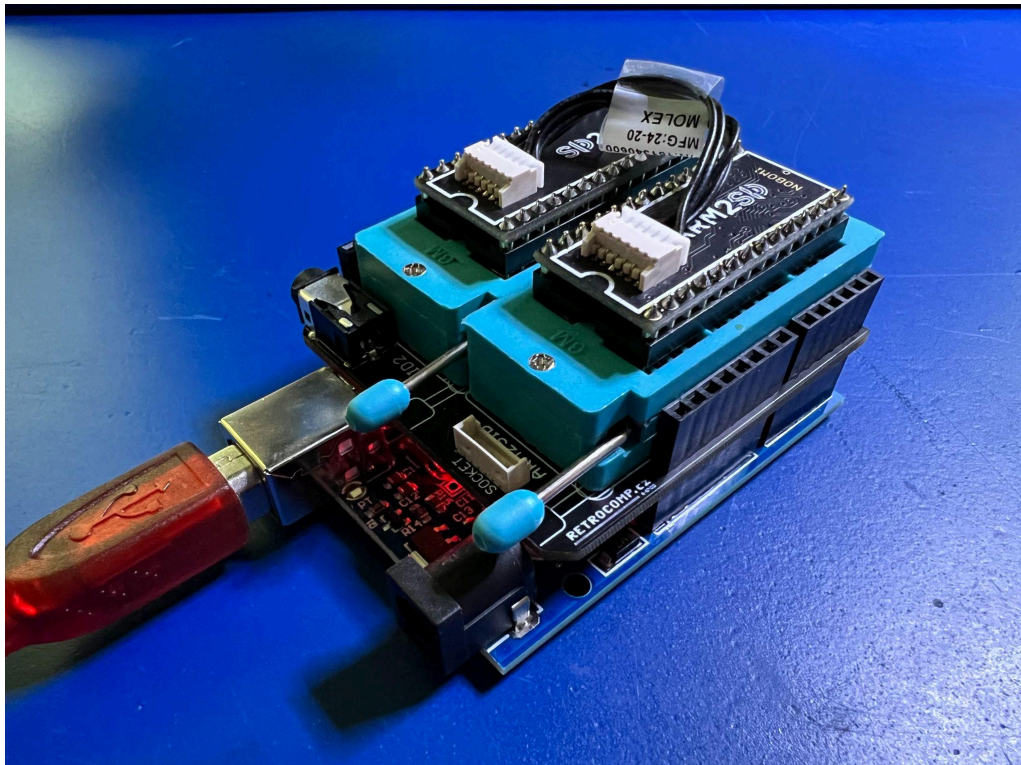
## Step 1 -Getting Started...

- Buy some ARMSIDs or ARM2SIDs and the ARMSID Shield from the Retrocomp.cz shop (sold with and without a parts BOM and sockets) -> <https://retrocomp.cz/produkt?id=67>
- Download or clone the latest Github Repo for the ARMSID Configurator:  
<https://github.com/nobomi/Arduino-ARMSID-configurator>

- When your package arrives, assemble the ARMSID Shield, following the included build guide



- Make sure it fits properly onto your Arduino Uno board



## Step 2 - Programming the Uno Board with the Arduino IDE

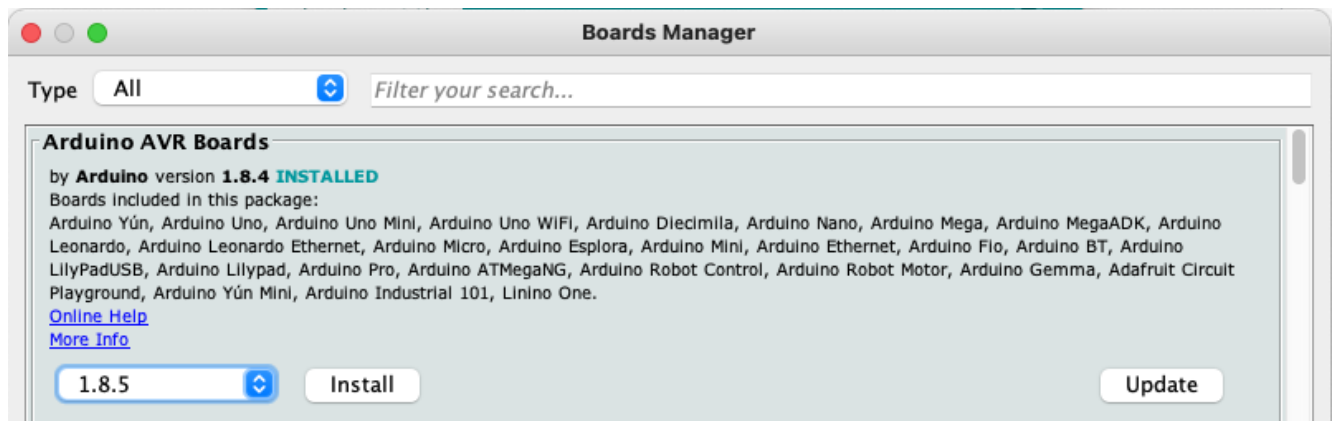
Be aware that the ARMSID Configurator requires multiple, different, microcontroller software modules, in the form of Arduino sketches. You'll need to be sure to program the correct one to your Uno board to manage the function(s) you want to perform.

You'll use the "**ARMSID\_test.ino**" sketch to test your ARMSIDs and/or to modify their internal Firmware settings

You'll use the "**ArduinoUpdater.ino**" sketch to load a new Firmware version to your ARMSIDs

You'll use the "**ArduinoPlay.ino**" sketch to test & play SID music directly from your ARMSID Shield, which you can hear by using headphones with 3.5mm plug.

- Before getting started, be sure to have the latest Arduino IDE downloaded & installed  
<https://www.arduino.cc/en/software>
- You'll also need to use the Arduino Boards Manager to download and install the latest software for AVR Boards

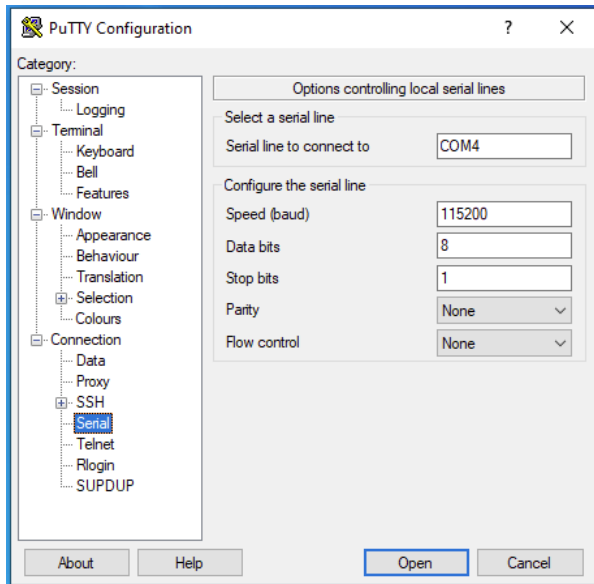


## Step 3 - Use an ANSI Terminal to Interact with your ARMSIDs

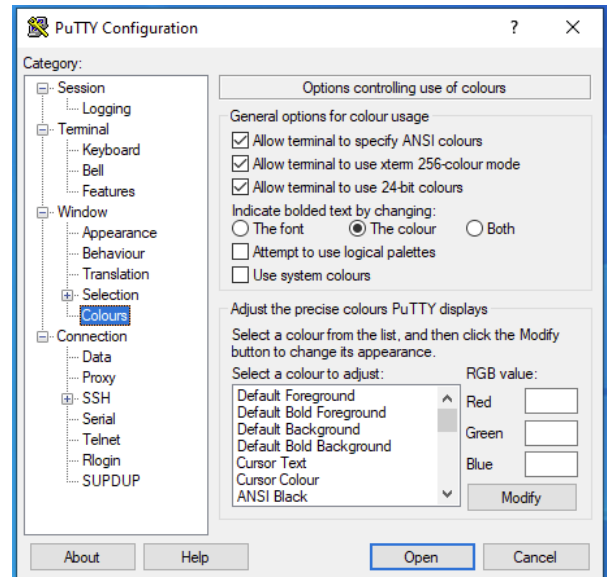
First, ensure that your Arduino Uno has been flashed with the required **ARMSID\_test.ino** sketch. Next, you should *always* use a Terminal program that provides **full** ANSI support, which will support control signals, menu navigation, graphics characters, colors, etc. If you try to use a less capable shell or Terminal, like the Mac terminal, or another CLI based Terminal emulator, you will get mixed results (*save yourself the headache*).

Using **PuTTY** to interact with the Uno to Test and configure the ARMSID firmware.

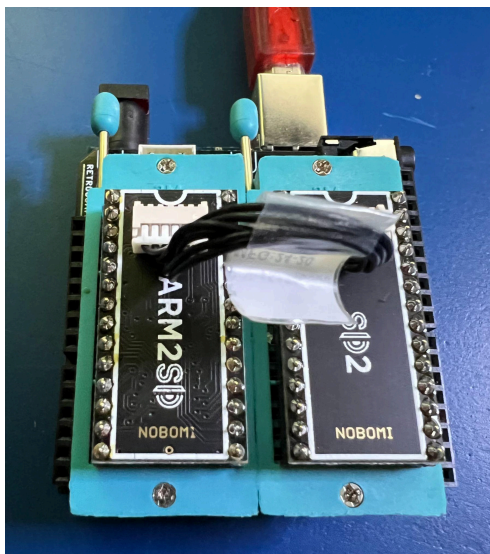
### Use Serial @ 115200, 8N1 Settings



### Configure for ANSI Color Controls

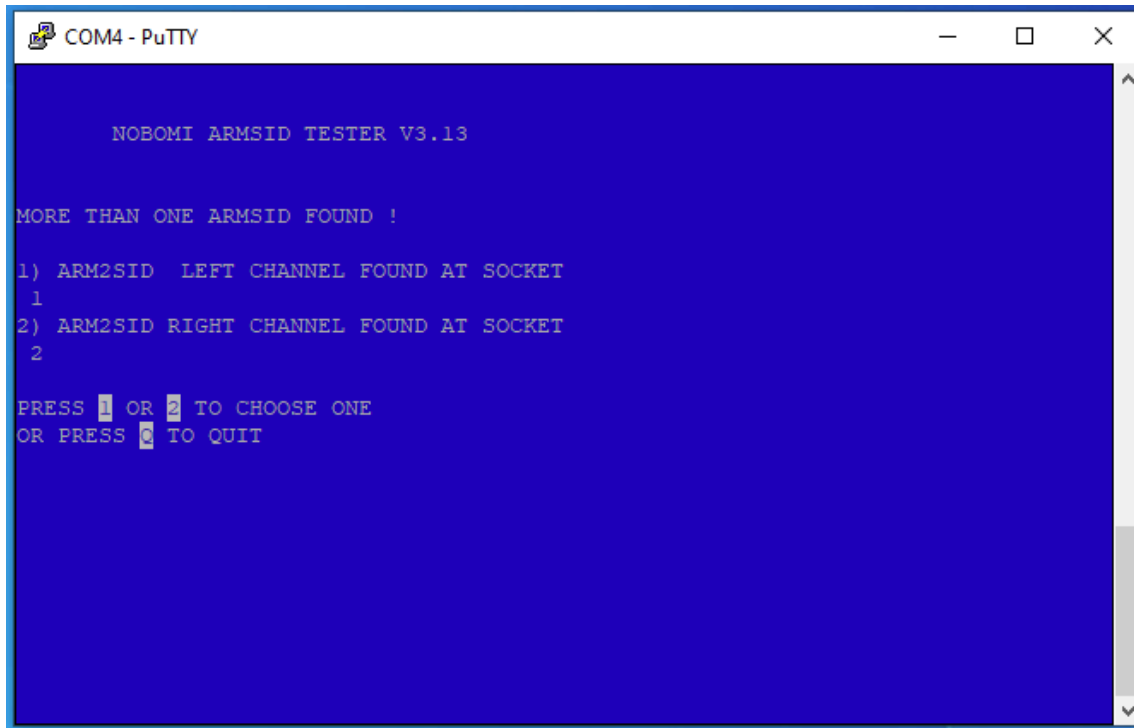


- Ensure that the primary ARM2SID is in the left Socket, and the “dummy” ARM2SID (labeled SID2) is in the right Socket





- Establish a Serial connection to the Uno
- Press “Enter” a couple times to bring up the main menu



```

COM4 - PuTTY

NOBOMI ARMSID TESTER V3.13

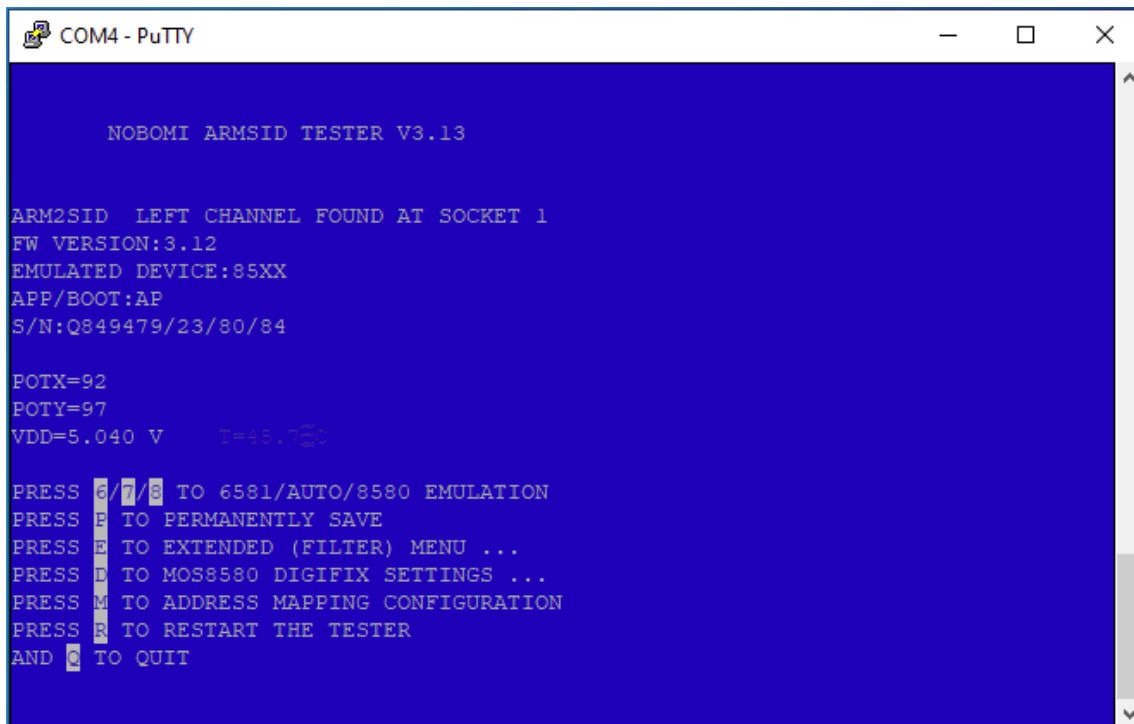
MORE THAN ONE ARMSID FOUND !

1) ARM2SID LEFT CHANNEL FOUND AT SOCKET
  1
2) ARM2SID RIGHT CHANNEL FOUND AT SOCKET
  2

PRESS 1 OR 2 TO CHOOSE ONE
OR PRESS 0 TO QUIT

```

- Select 1 to program the LEFT Channel



```

COM4 - PuTTY

NOBOMI ARMSID TESTER V3.13

ARM2SID LEFT CHANNEL FOUND AT SOCKET 1
FW VERSION:3.12
EMULATED DEVICE:85XX
APP/BOOT:AP
S/N:Q849479/23/80/84

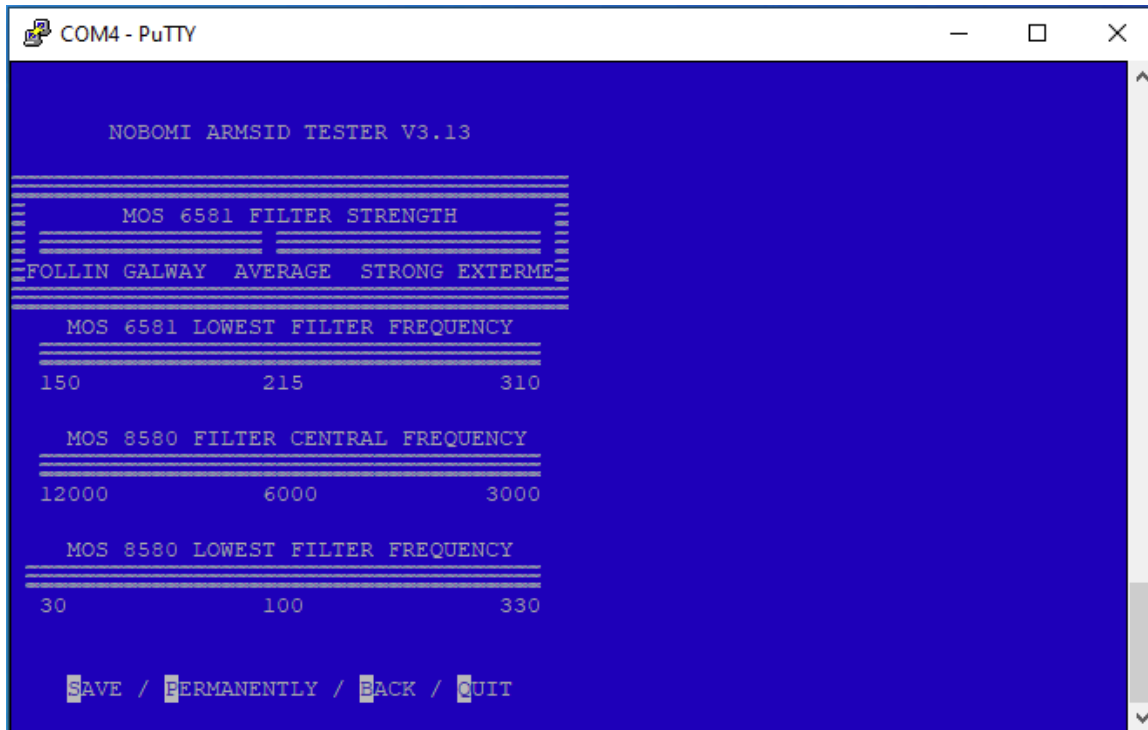
POTX=92
POTY=97
VDD=5.040 V T=48.7°C

PRESS 6/7/8 TO 6581/AUTO/8580 EMULATION
PRESS P TO PERMANENTLY SAVE
PRESS E TO EXTENDED (FILTER) MENU ...
PRESS D TO MOS8580 DIGIFIX SETTINGS ...
PRESS M TO ADDRESS MAPPING CONFIGURATION
PRESS R TO RESTART THE TESTER
AND 0 TO QUIT

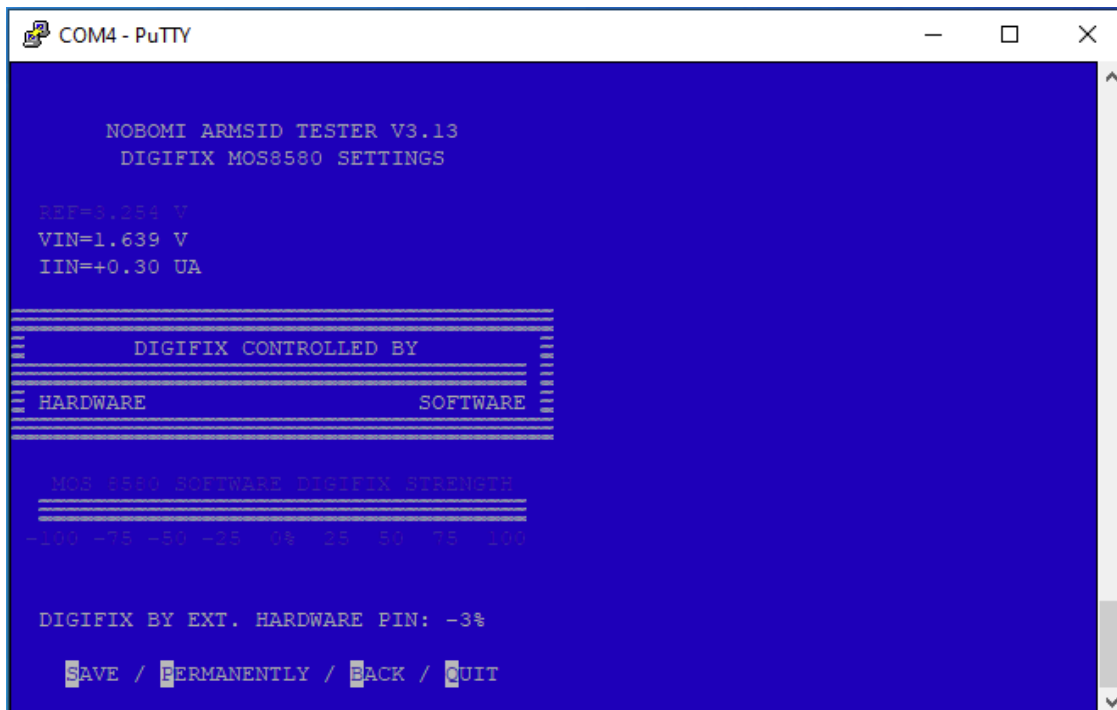
```

- Press 6 to set your ARMSID for 6581 SID Emulation, or 8 for 8580
- Press P to Permanently save this setting to the ARMSIDs internal memory
- Press E for the Extended Filter Menu

- Use the UP/DOWN arrow keys to move between menus and the LEFT/RIGHT arrow keys to select desired filter settings. For defaults, all settings should be in the middle, with the exception of the 8580 filter (bottom menu) which should be closer to 30Hz. When selections are done, press **S** to save preferences and **B** to return to the main menu



- Select **D** for 8580 Digifix settings
- Ensure this is set to **Hardware** as the MB6582 doesn't play Digi music. Press **S** to save and **B** to return to the main menu



- Press **M** for Address Mapping
- Ensure Pin Connections is set to **SOCKET** as opposed to Wire and SID/1 is set to **SIDL** and SID/2 is set to **SIDR**. Press **S** to save and **B** to return to the main menu



- Press **P** to Permanently save all settings
- Press **R** to restart the tester
- Press **2** to select the RIGHT socket
- Repeat all of the same menu selections and settings for the RIGHT Socket, then permanently save again

## Step 4 - Load new Firmware into the ARMSID

First, ensure that your Arduino Uno has been flashed with the required **ArduinoUpdater.ino** sketch. Next, place the desired firmware that you want to flash to your ARMSID into the same folder as the ARMSIDupArduino.exe executable.

- Place your ARM2SIDs into the correct socket positions (see beginning of doc)
- From a Windows PC Command Prompt, run the updater using the following syntax:  
**ARMSIDupArduino.exe COMx a2sid314.bin**
- Press **Y** when prompted to perform the firmware update
- Example output below:

```
Opening serial port...OK
```

```
=====
```

```
NOBOMI ARM2SID UPDATE TO V3.14
```

```
=====
```

```
NOBOMI ARM2SID UPDATE TO V3.14
```

```
ARM2SID FOUND IN ARDUINO
```

```
FW VERSION:3.12
```

```
EMULATED DEVICE:85XX
```

```
APP/BOOT:AP
```

```
READY TO UPDATE?[Y/N]
```

```
Z:\Documents\Tech_Stuff\Electronics\Commodore Projects\C64 SID Clone - ARMSID
```

```
Configurator\Arduino\ARMSID_FW_updater\ARMSIDupdaterArduino>ARMSIDupArduino.exe COM4 a2sid314.bin
```

```
Opening serial port...OK
```

```
=====
```

```
NOBOMI ARM2SID UPDATE TO V3.14
```

```
=====
```

```
NOBOMI ARM2SID UPDATE TO V3.14
```

```
ARM2SID FOUND IN ARDUINO
```

```
FW VERSION:3.12
```

```
EMULATED DEVICE:85XX
```

```
APP/BOOT:AP
```

```
READY TO UPDATE?[Y/N]
```

```
SFX SWITCHING OFF .... DONE
```

```
=====
```



NOBOMI ARM2SID UPDATE TO V3.14

ARM2SID FOUND IN ARDUINO  
FW VERSION:3.12  
EMULATED DEVICE:85XX  
APP/BOOT:AP

=====

NOBOMI ARM2SID UPDATE TO V3.14

ARM2SID FOUND IN ARDUINO  
FW VERSION:3.12  
EMULATED DEVICE:85XX  
APP/BOOT:AP

ERASING APP HEADER ... DONE

RESETTING ... DONE

=====

NOBOMI ARM2SID UPDATE TO V3.14

ARM2SID FOUND IN ARDUINO  
FW VERSION:3.12  
EMULATED DEVICE:85XX  
APP/BOOT:BL

ERASING APP ..... DONE

BLANK CHECK ... OK

PROGRAMMING ... DONE

RESETTING ....DONE

=====

NOBOMI ARM2SID UPDATE TO V3.14

ARM2SID FOUND IN ARDUINO  
FW VERSION:3.14  
EMULATED DEVICE:85XX  
APP/BOOT:AP

UPDATE DONE

## Step 5 - Play sample SID tunes to test your ARMSID Configuration

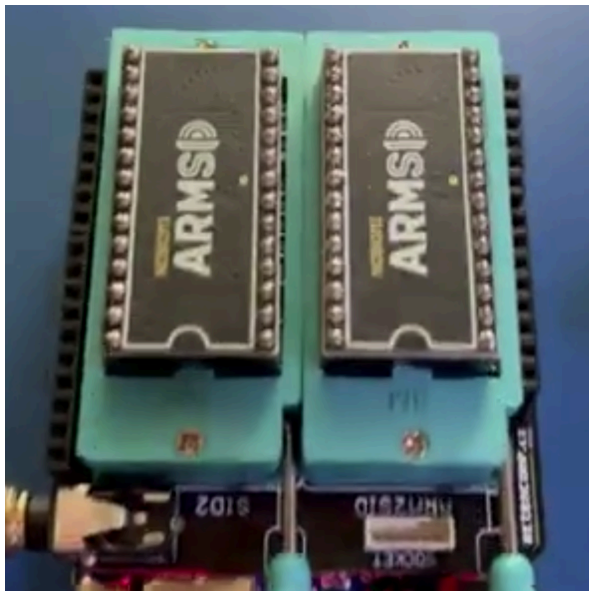
First, ensure that your Arduino Uno has been flashed with the required **ArduinoPlay.ino** sketch.

- Place your ARM2SIDs into the correct socket positions (see beginning of doc)
- Plug a set of headphones into the 3.5mm audio jack on the ARMSID Shield
- From a Windows PC Command Prompt, run the updater using the following syntax:  
**SIDplay410.exe comx *Hexadecimal\_2SID.sid***
- Example output below:

```
Opening serial port...OK
```

[illegible]

**Click on the image below to link to a testing video:**



<https://youtu.be/8GP55YnrpcY>

## Step 6 - Add the ARMSIDs into your MIDIBox Synth

The MIDIBox MB6582 is an amazing little synth, supporting for up to 4x SID Engines, with each providing 2x SIDs in a stereo configuration, for rich textured and layered chiptune glory. The MB6582 design was put together by Jason Williams (aka “Wilba”), using a combination of MIDIBox modules made available by Thorsten Klose, who founded the MIDIBox community. The passion to have one of these amazing instruments in the collection has driven many folks (like myself) to learn “do it yourself electronics” (DIY), since that’s typically the only way to get ahold of one of these.

[http://www.midibox.org/dokuwiki/doku.php?id=wilba\\_mb\\_6582](http://www.midibox.org/dokuwiki/doku.php?id=wilba_mb_6582)



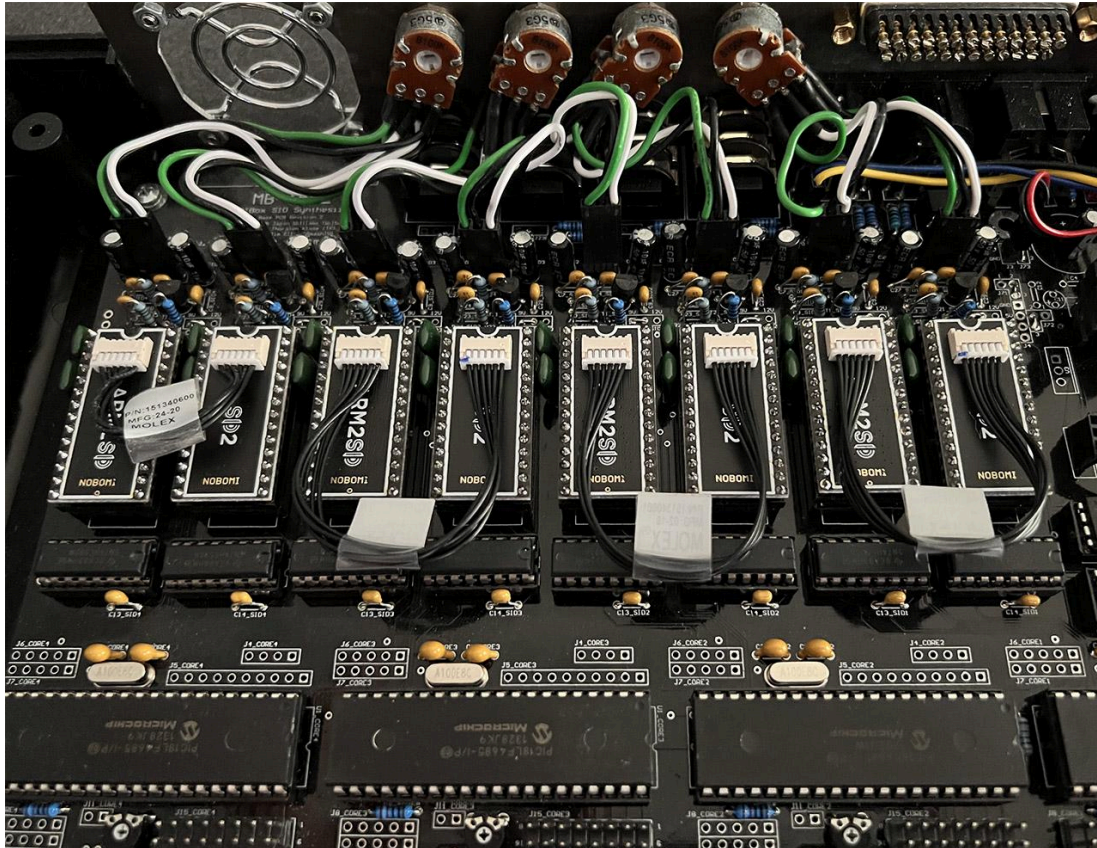
Needless to say, having to source 8x properly working and perfect sounding original MOS SID chips is getting ever closer to becoming a losing proposition. As the available supply becomes thinner and thinner, the market becomes riskier and riskier with price-gougers, content to defraud potential buyers or withhold information about faulty filters and failed oscillators, etc.

Using ARMSIDs in place of the originals provides you with a readily available, modestly priced and amazing sounding alternative, that lets you just get on with your build, worry-free.

There are only a few considerations you'll want to take into consideration when using ARMSID in your synth.

### Proper Placement

- The MB6582's SID Engines are deployed from RIGHT to LEFT, meaning you should start populating your ARMSIDs into the last two SID sockets on the RIGHT-most side of MB6582's base PCB first
- The primary ARM2SID (labeled accordingly) should go on the left side of the SID module, and the dummy ARMSID (labeled SID2) should go into the right socket of each SID module
- Be sure to connect the Molex cables between your ARM2SID pairs



### Sound Configuration Preferences

- Given that the ARM2SIDs use DSP code to emulate the sounds of the SID, you *may* want to consider tweaking some of the ARMSID configuration parameters during the firmware setup. This will introduce sound differences between the left and right pairs, which can be pleasing to hear in stereo
- You could make subtle differences in the configurations, like introducing offsets in the filter settings for each chip in ARM2SID pair, or more extreme differences like making one chip sound & behave like a 6581, with all of its character, while making the other respond like an 8580
- ***The choice is yours!***

### Considerations

- ARMSIDs do not care about the SIDs VDD voltage levels (Pin 28), so a 12v or 9v setting will make no difference. The ARMSID will draw its power from the 5v VCC (Pin 25)
- In fact, if you're making an MB6582 that will *only* ever use ARMSIDs, you won't require a 12v / 9v voltage rail at all, which can simplify your build substantially
- ARMSIDs will not use the SIDs filter capacitors, so you can leave these unpopulated as well